

Android Studio

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Code 1: Context Menu

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:padding="16dp"
        android:text="Contact List"
        android:textSize="20sp"
        android:textColor="@android:color/white"
        android:background="@color/purple_500" />

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item
        android:id="@+id/action_call"
        android:title="Call"
        android:icon="@drawable/menu_call" />

    <item
        android:id="@+id/action_email"
        android:title="Email"
        android:icon="@drawable/ic_mail" />

    <item
        android:id="@+id/action_edit"
        android:title="Edit Contact" />

    <item
        android:id="@+id/action_delete"
        android:title="Delete Contact" />

</menu>
```

```

package com.example.project1java;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.ContextMenu;
import android.view.MenuItem;
import android.view.View;
import android.widget.AdapterView;
import android.widget.AdapterView.OnItemClickListener;
import android.widget.ArrayAdapter;
import android.widget.EditText;
import android.widget.ListView;
import android.widget.Toast;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;

import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    ListView listView;
    ArrayAdapter<String> adapter;
    ArrayList<Contact> contactList;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        listView = findViewById(R.id.listView);

        contactList = new ArrayList<>();

        contactList.add(new Contact("Amit Sharma", "amit@gmail.com",
"9876511111"));
        contactList.add(new Contact("Pooja Verma", "pooja@gmail.com",
"9876543210"));
        contactList.add(new Contact("Rahul Mehta", "rahul@gmail.com",
"9876543212"));
        contactList.add(new Contact("Sneha Gupta", "sneha@gmail.com",
"9876543333"));
        contactList.add(new Contact("Neha Kapoor", "neha@gmail.com",
"9876544444"));

        adapter = new ArrayAdapter<>(
            this,
            android.R.layout.simple_list_item_1,
            getNames());

        listView.setAdapter(adapter);

        // Register context menu for ListView
        registerForContextMenu(listView);
    }

    private ArrayList<String> getNames() {

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        ArrayList<String> names = new ArrayList<>();

        for (Contact c : contactList) {
            names.add(c.name + "\n" + c.email);
        }

        return names;
    }

    // Create the menu from XML file
    @Override
    public void onCreateContextMenu(
        ContextMenu menu,
        View v,
        ContextMenu.ContextMenuInfo menuInfo) {

        super.onCreateContextMenu(menu, v, menuInfo);

        getMenuInflater().inflate(R.menu.contact_menu, menu);
    }

    // Handle menu item selection
    @Override
    public boolean onContextItemSelected(MenuItem item) {

        AdapterView.AdapterContextMenuInfo info =
            (AdapterView.AdapterContextMenuInfo) item getMenuInfo();

        int position = info.position;
        Contact selectedContact = contactList.get(position);

        if (item.getItemId() == R.id.action_call) {

            makePhoneCall(selectedContact);
            return true;

        } else if (item.getItemId() == R.id.action_email) {

            sendEmail(selectedContact);
            return true;

        } else if (item.getItemId() == R.id.action_edit) {

            showEditDialog(selectedContact);
            return true;

        } else if (item.getItemId() == R.id.action_delete) {

            showDeleteDialog(position);
            return true;

        }

        return super.onContextItemSelected(item);
    }

    // Call contact

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private void makePhoneCall(Contact contact) {

    Intent intent = new Intent(
        Intent.ACTION_DIAL,
        Uri.parse("tel:" + contact.phone)
    );

    startActivity(intent);
}

// Send email
private void sendEmail(Contact contact) {

    Intent intent = new Intent(
        Intent.ACTION_SENDTO,
        Uri.parse("mailto:" + contact.email)
    );

    startActivity(intent);
}

// Edit contact
private void showEditDialog(Contact contact) {

    EditText editText = new EditText(this);
    editText.setText(contact.name);

    new AlertDialog.Builder(this)
        .setTitle("Edit Contact")
        .setMessage("Enter new contact name")
        .setView(editText)
        .setNegativeButton("CANCEL", null)
        .setPositiveButton("SAVE", (dialog, which) -> {

            contact.name = editText.getText().toString();

            adapter.clear();
            adapter.addAll(getNames());
            adapter.notifyDataSetChanged();

            Toast.makeText(
                this,
                "Contact updated",
                Toast.LENGTH_SHORT
            ).show();
        })
        .show();
}

// Delete contact
private void showDeleteDialog(int position) {

    Contact contact = contactList.get(position);

    new AlertDialog.Builder(this)
        .setTitle("Delete Contact")
        .setMessage("Are you sure you want to delete " + contact.name

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+ "?")
        .setNegativeButton("CANCEL", null)
        .setPositiveButton("DELETE", (dialog, which) -> {

            contactList.remove(position);

            adapter.clear();
            adapter.addAll(getNames());
            adapter.notifyDataSetChanged();

            Toast.makeText(
                this,
                "Contact deleted",
                Toast.LENGTH_SHORT
            ).show();
        })
        .show();
    }

    // Contact class
    static class Contact {

        String name;
        String email;
        String phone;

        Contact(String name, String email, String phone) {
            this.name = name;
            this.email = email;
            this.phone = phone;
        }
    }
}

```